

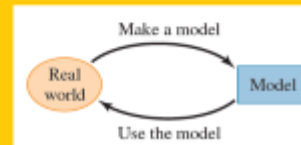
SSILP Math

Beginner

MATH MODELING &
BRIDGE BUILDING ACTIVITY



Math Modeling



Design a Paper Bridge using Math modeling

<Group Activity: 1.5 hours>

This activity is all about seeing how the math we learn in class can be applied in real-world engineering and architecture.

Have fun, get creative, and let's build some amazing bridges together!

Learning objectives

Understand Mathematical Modeling: Students will learn what a mathematical model is and how it can be used to represent real-world objects or processes

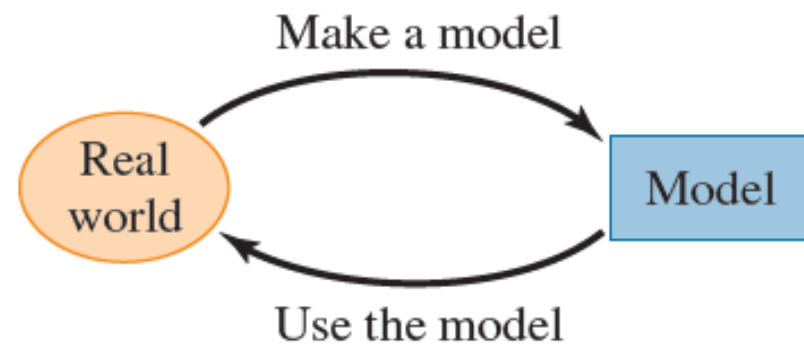
Apply Regression Analysis: Students will understand how to create scatter plots and regression lines using Excel, and how to interpret the slope of the regression line

Design and Build a Paper Bridge: Students will engage in a group activity to design and construct a paper bridge using mathematical modeling principles

Test and Analyze Bridge Performance: Students will conduct load testing on their bridges, record data, and analyze the results using scatter plots and regression analysis

Present Findings: Students will present their bridge designs, testing results, and comparisons of different designs to the class

Math Modeling



What is a model?

A **model** is a representation of an object or process.

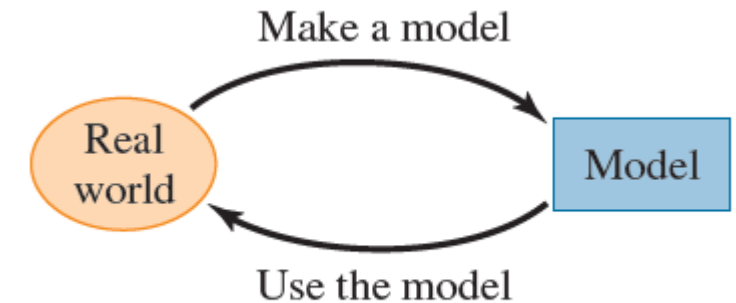
For example, a toy Ferrari is a model of the actual car

A road map is a model of the streets in a city.

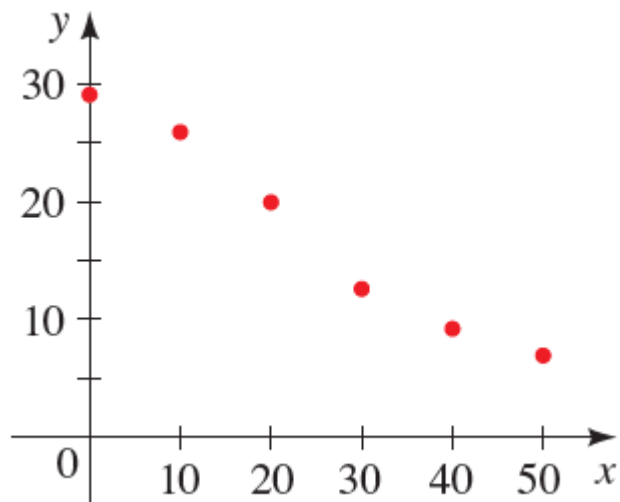
Architects make models of buildings and cities before they are actually built.

A mathematical model is a **mathematical representation** (usually an equation) **of an object or process**.

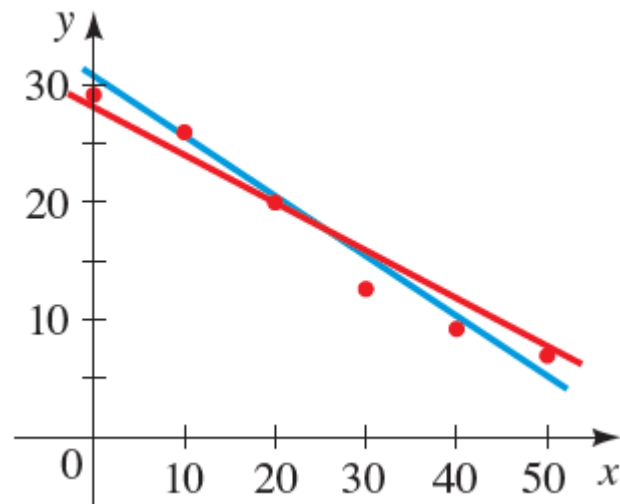
We use models to **obtain useful information** or **make predictions** about the thing being modeled.



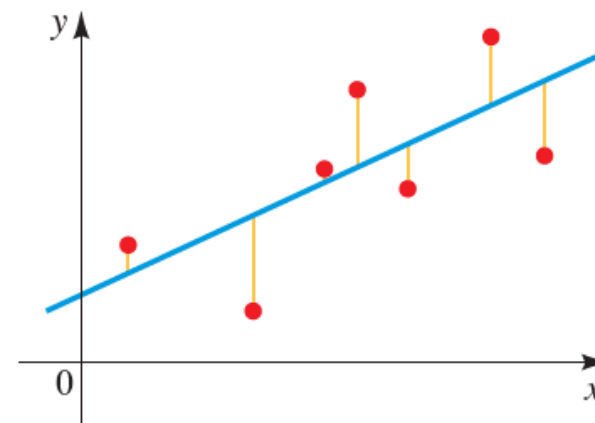
A regression line



U.S. infant mortality rate

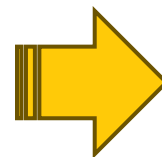


Visual attempts to fit line to data



Which is the **line that best fits** the data?

→ It is the line that is as close as possible to all the data points.

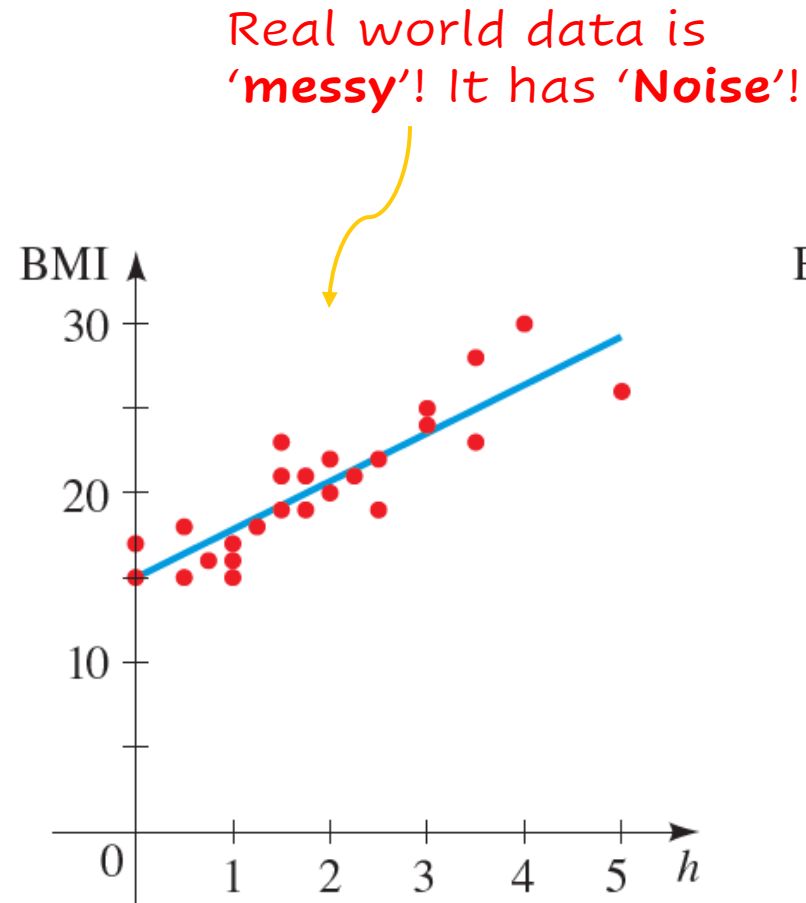


A Regression Line

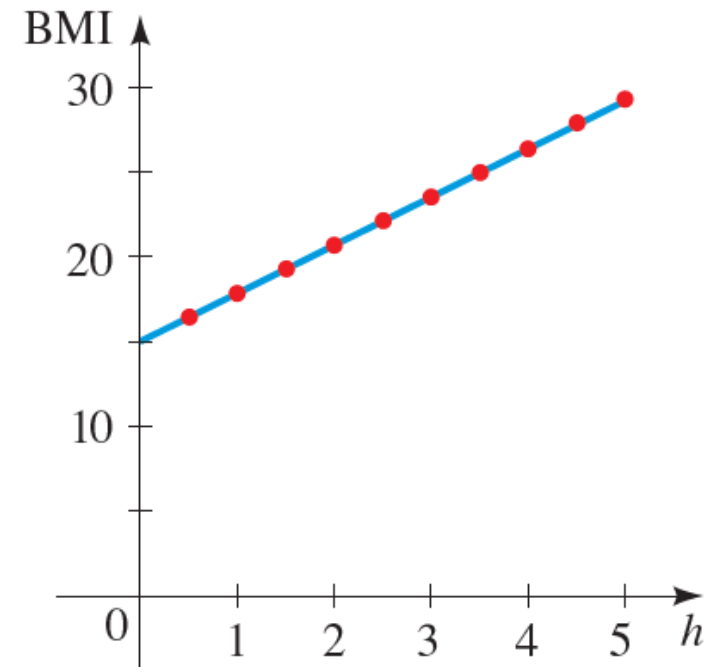
This is the line for which the **sum of squares of the vertical distances** from the data points to the line is as **small** as possible

A Line That Best Fits the Data

A scatter plot



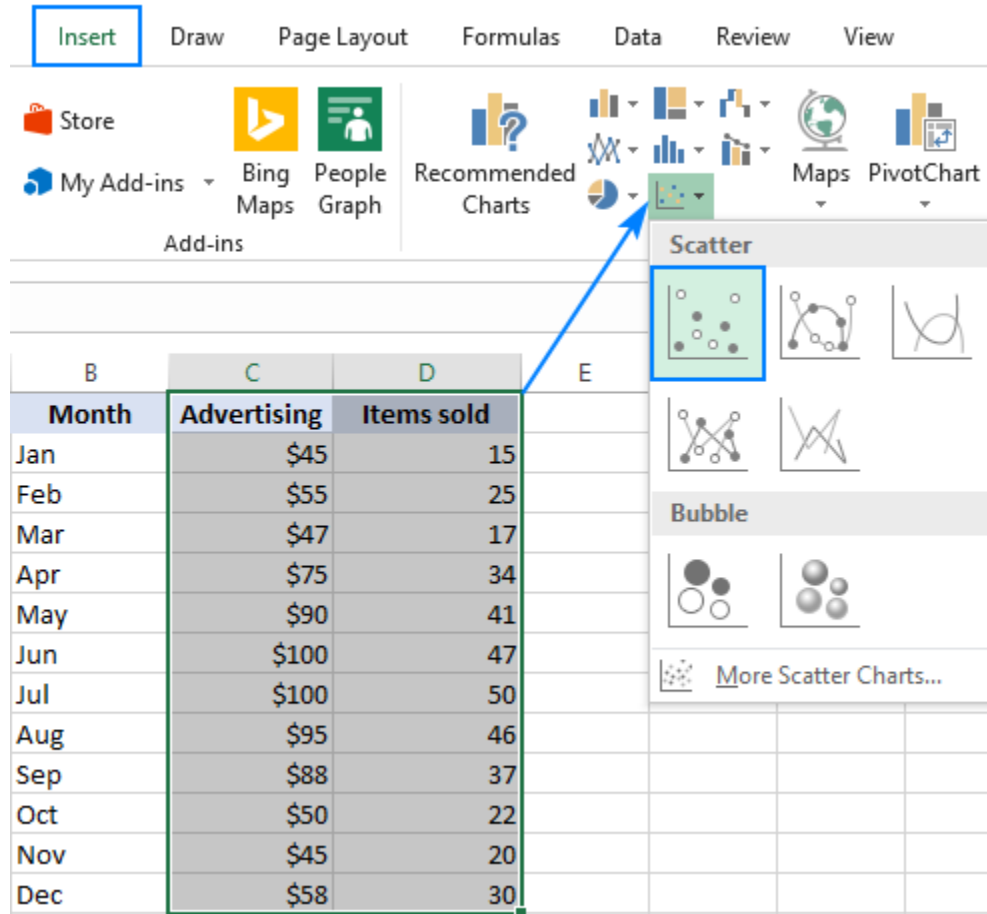
(a) Line of best fit



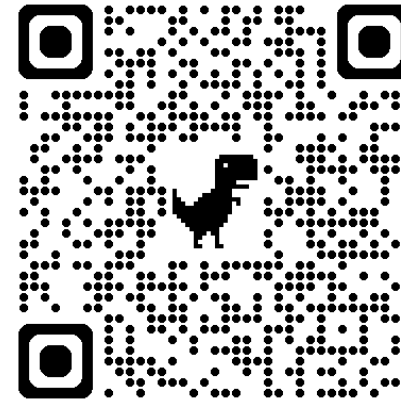
(b) Line fits data exactly

We can see a linear trend

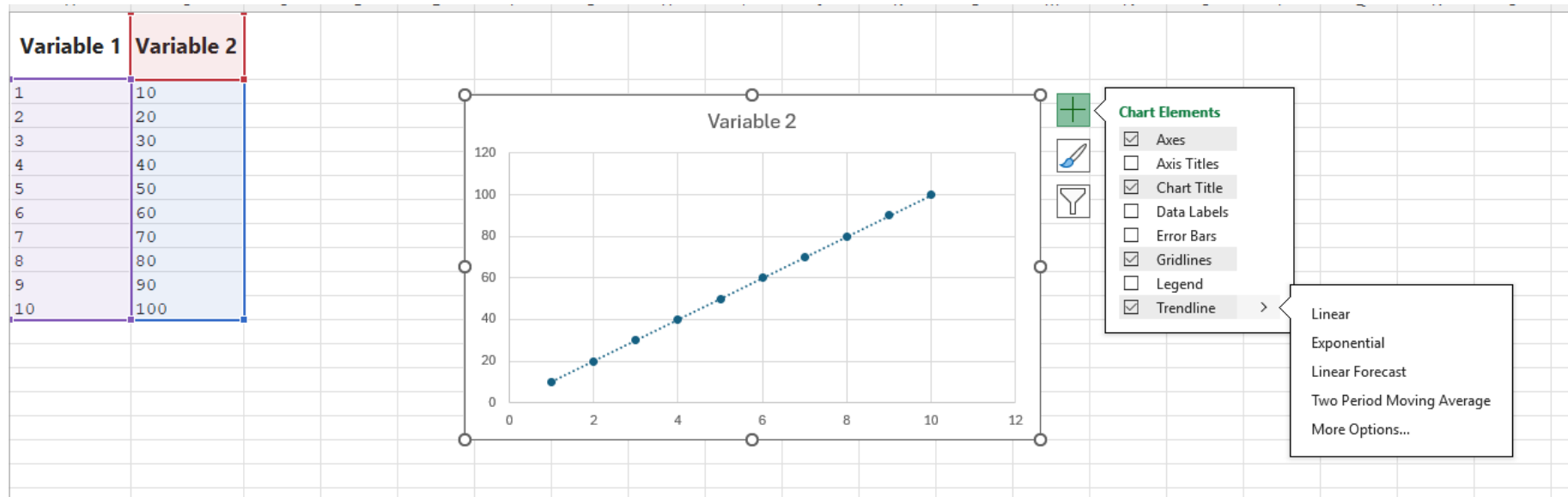
How to create a scatter plot in excel



<https://www.ablebits.com/office-addins-blog/make-scatter-plot-excel/>



How to create a regression line using excel

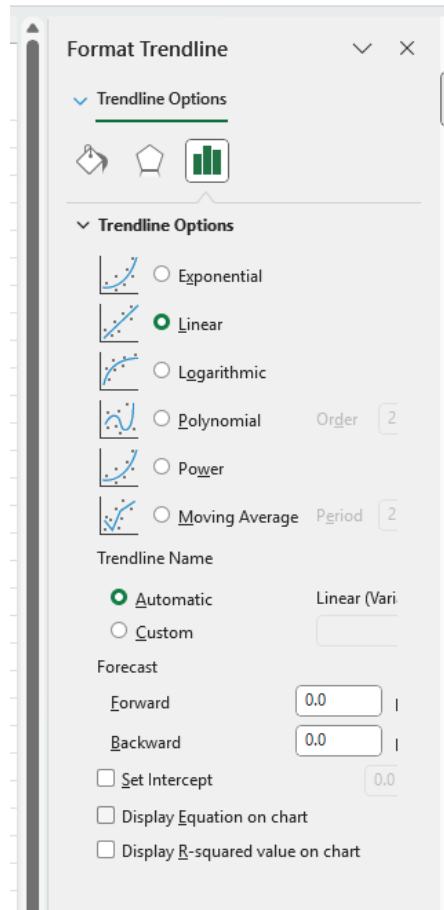


1.Ensure Data is Plotted in a scatter plot.

2.Add a Trendline:

1. Select the chart ("Chart 1").
2. Right-click on any data point in the chart and choose **Add Trendline**.
3. In the **Format Trendline** pane, select the type of trendline (e.g., Linear for regression).

How to display line equation on the chart



- To add the line equation, under the “Trendline” menu select “More Trendline Options” to see the “Format Trendline” window shown below. Select the boxes next to “Display equation on chart” and you are all set.

Design a Paper Bridge using Math modeling

<Group Activity: 1.5 hours>

This activity is all about seeing how the math we learn in class can be applied in real-world engineering and architecture.

Have fun, get creative, and let's build some amazing bridges together!

Step 1: Understand

Understanding Variables and Functions:

First, let's think about the different aspects of a bridge that you might want to measure and model. Here are a few ideas to get you started, but feel free to come up with your own

Variables to consider:

Span (s): Distance between supports

Depth (d): Beam height

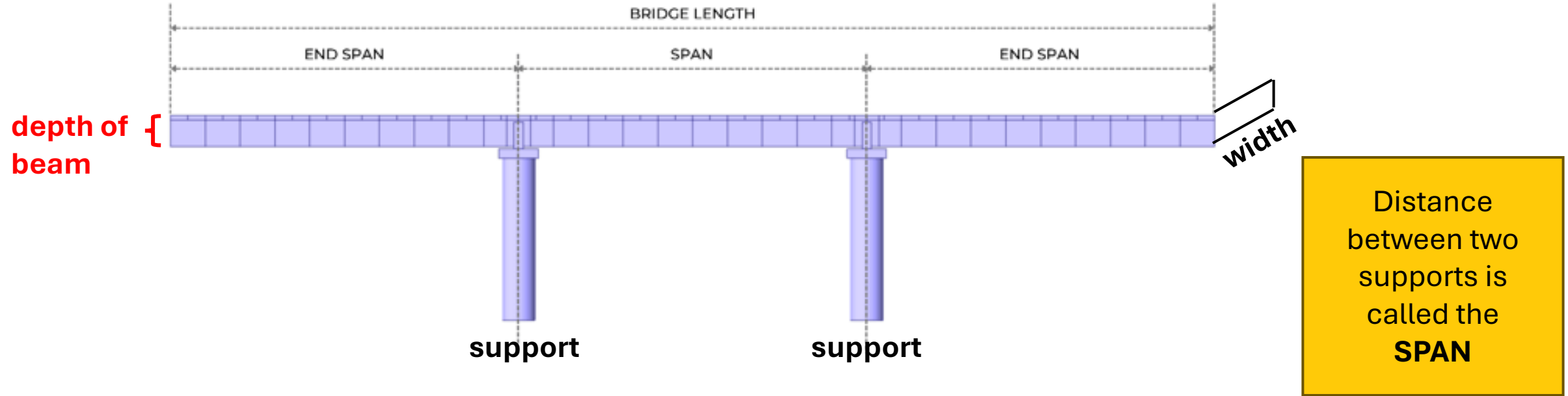
Width (w): Bridge width

Number of horizontal beams (b)

Number of supports (n): Single or double span

Weight (L): Max load before failure

Quantity of material (m):



For a given span, the **depth of the beam** will determine how rigid the bridge is.

If it's deeper it will be stronger **but** will **use more material and be heavier**

Step 2: Brainstorm

Form groups and brainstorm your bridge design. Sketch some options on paper & decide which idea you can pursue in the given time.



Step 3: Build

Constructing Your Bridge:

Grab your paper and start building! Use your sketches as a guide and be ready to adjust your design as you go.

- How high can you go?
- How strong can it be?
- How light can it be (saves materials!) while being able to support a given load?

Keep track of your variables and how changes affect your design. Use your creativity and engineering skills to bring your model to life!

Step 4: Modeling & representation

Types of Functional Relationships you can model:

Linear: Load vs. depth (initial approximation)

Inverse: Load vs. span (example $L \propto 1/s$)

Quadratic: Load vs. number of layers or width in some setups

Piecewise or nonlinear: For layered beams or truss-like designs

Think about:

- What symbols will you use to represent the variables?
- What are the relationships between the variables? (eg, *How much load can the bridge take with decrease with width?*)
- Which mathematical functions might best describe the relationships between these variables?

Step 4: Modeling & representation

TASK:

1. **Vary one variable at a time** and record load-bearing capacity.
2. **Graphing**: Use scatter plots for each variable pair and try different regression types (linear, quadratic).

Example: With the table below, a team can document the load strength of their bridge (output, dependant variable) based on the layers in the bridge (input, independent variable)

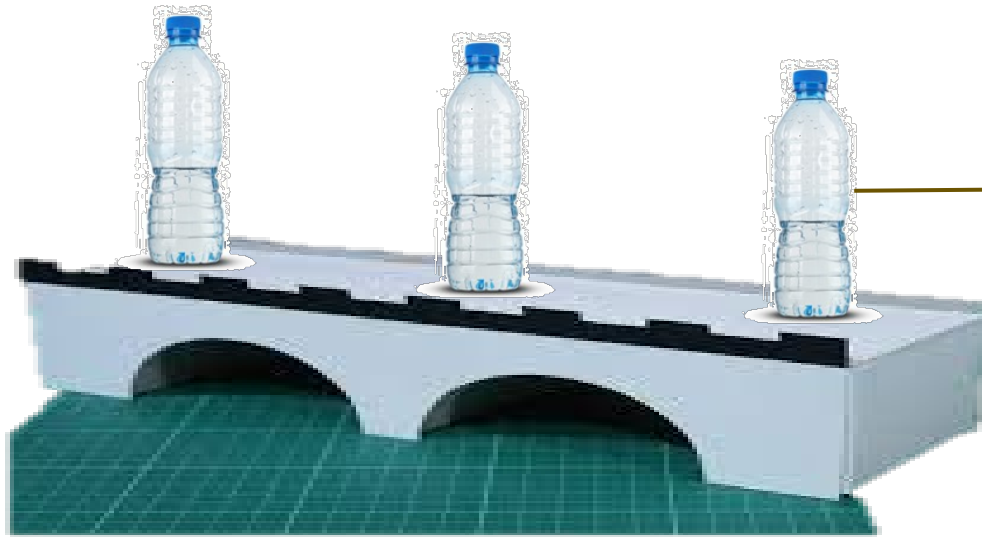
Layers in bridge	0	1	2	3	4	5	6
Load strength	0						

Step 5 : Testing and Analysis

Once your bridge is built, it's time to test it!

1. **Load Testing:** Gradually add weight to see how much load your bridge can support with particular input variables. Note the maximum weight it holds before any failure.
2. Also note the weight of the bridge using the weighing scales (*How will you do that?*)

A high ratio of $\frac{\text{max load}}{\text{weight of bridge}}$ indicates a strong bridge

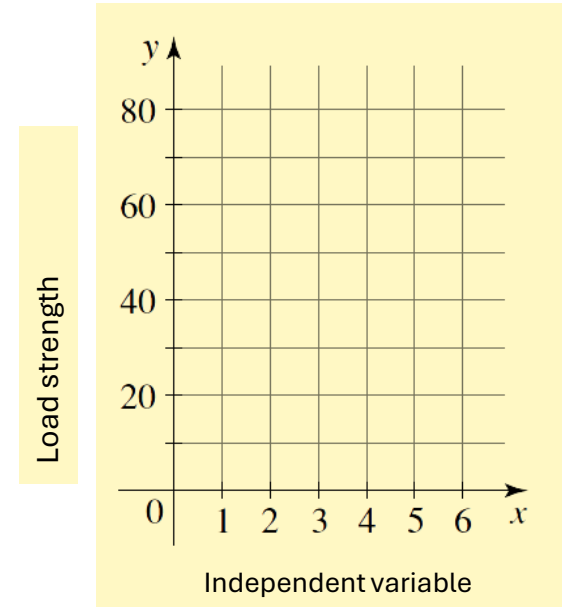


For load testing, you can use bottles of water with markings.

Increase the quantity of water gradually and systematically!

Step 5 : Testing and Analysis

1. Make a scatter plot of your data in Excel. Do the data follow an approximately linear trend?
2. Use Excel to find the regression line for your data.
3. What is the slope of the regression line, and what does it tell us?
4. Use the regression line to predict the load strength of a bridge with more independent variables

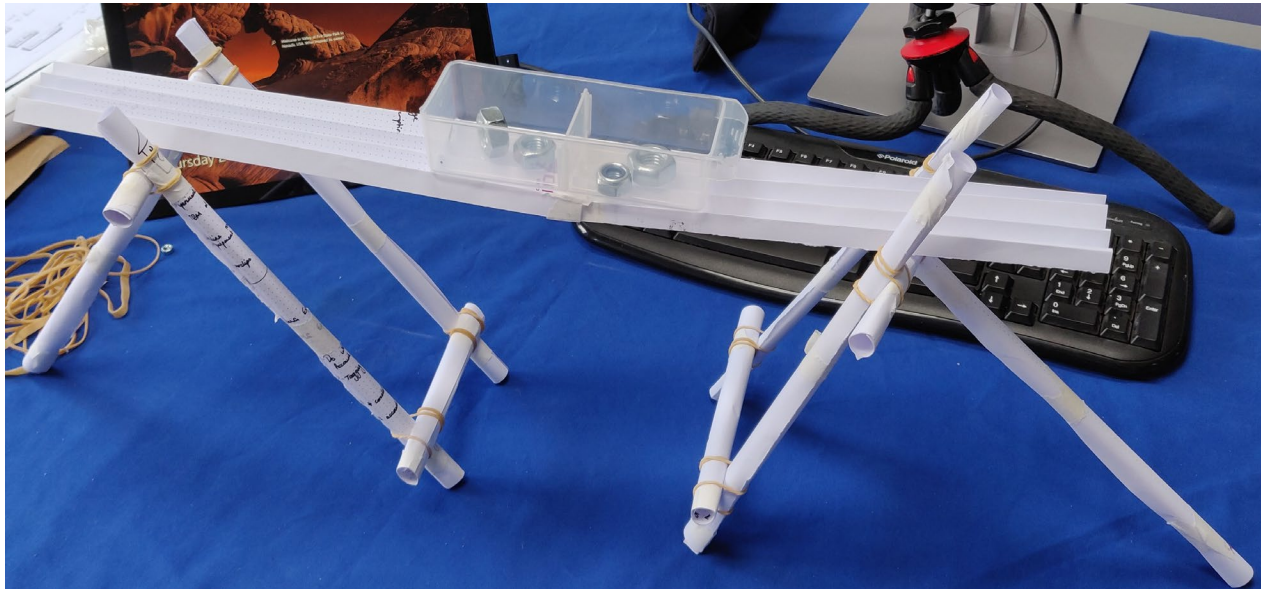
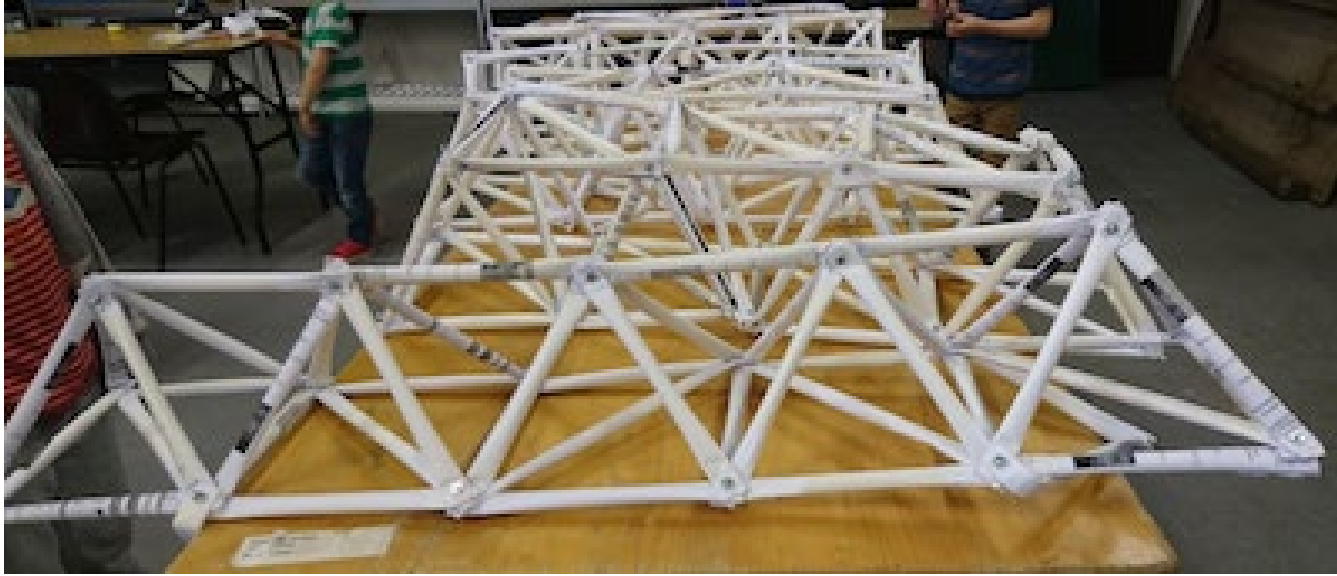


Step 6: Present your ideas!

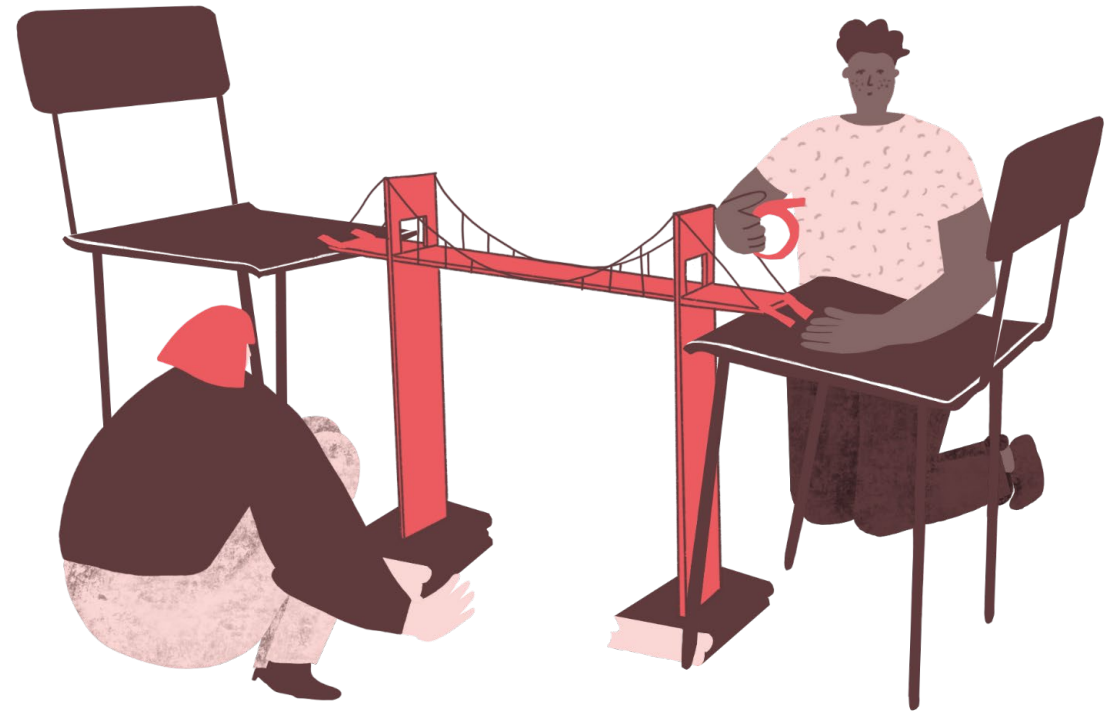
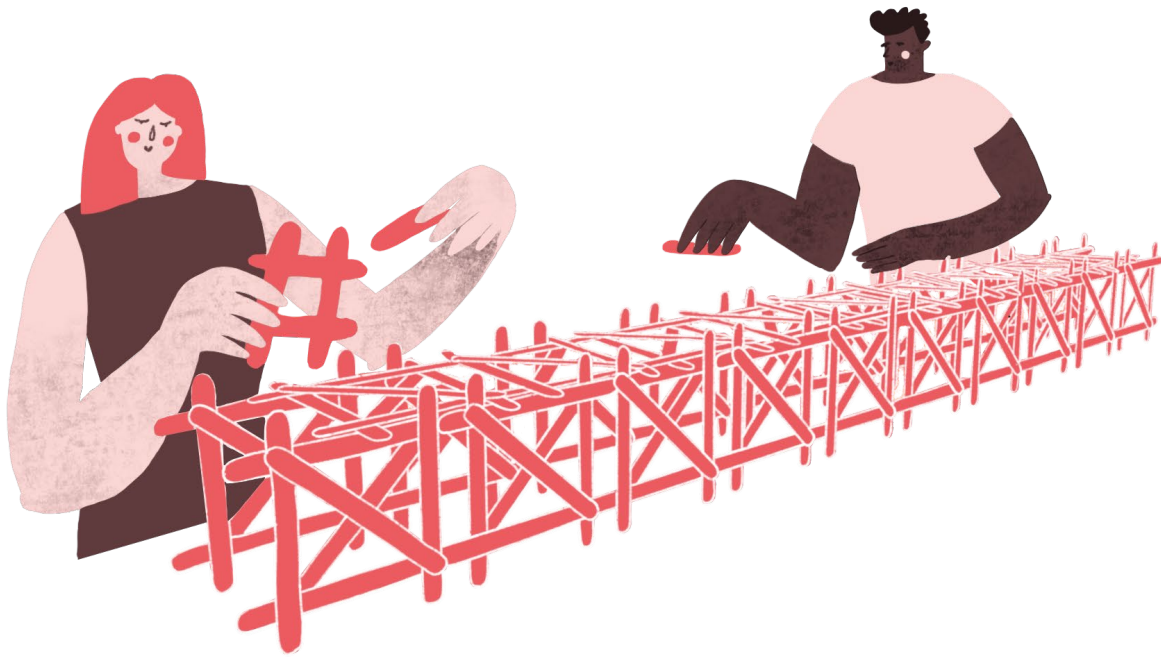
Each group will present their bridge to the class using 5 slides.

Here's what to cover in your presentation:

- **Design Explanation:** Describe your bridge design and the variables you chose to measure. Explain why you chose these variables and the functions you used to model them.
- **Testing Results:** Share how much load your bridge supported and how stable it was. Explain any adjustments you made during the building process.
- **Design Comparisons:** If you tested multiple designs, compare the results. Explain how different spans, heights, or support positions affected the performance.



Get
ambitious!



EXAMPLE

Submission

Team Submission on E-dimension

Submit your slide deck (5 slides) on Edimension.

Cover slide must include your

- Bridge name,
- team members names and student ids.

Only one team member per team needs to upload the slides

Deadline: by Monday of Week 4 (6 pm)